

TESTS ON AGGREGATES

Aggregate in Concrete



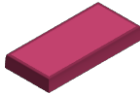
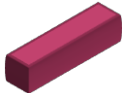
Source: PCA



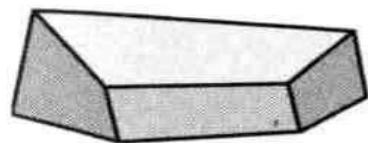
Aggregate Properties

- **Aggregates' properties are used to determine if:**
 - Suitability for a particular application
 - Types needed for specific concrete mix design
- **Following aggregate properties are considered:**
 - Shape and texture
 - Soundness and durability
 - Hardness and abrasion resistance
 - Absorption
 - Specific gravity
 - Strength
 - Gradation
 - Cleanness and deleterious materials
 - Alkali-aggregate reactivity

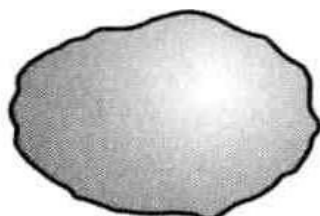
Aggregate Particle Shapes

- **Rounded:** river gravel 
 - Better packing, lowest voids ratio= 33%
 - Less interlocking between particles
- **Angular:** crushed rock 
 - Loose packing, higher voids ratio
 - Interlocking between particles is good
- **Flaky:** small thickness 
- **Elongated:** length considerable 
- **Flaky & elongated:** thin & long
 - Bad for concrete durability because of easy breakage and difficulty compacting
 - Should be restricted to 10-15% in concrete design

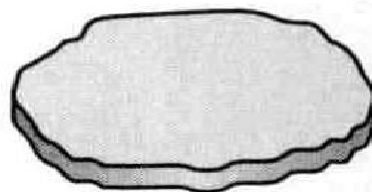
Aggregate Particle Shapes



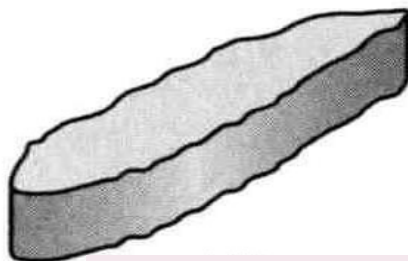
Angular



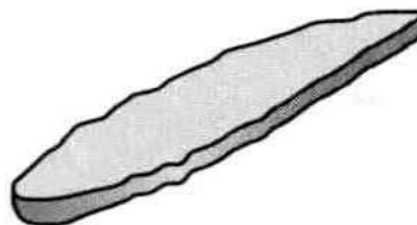
Rounded



Flaky



Elongated

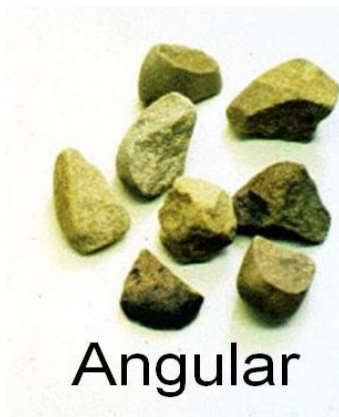


Flaky & Elongated

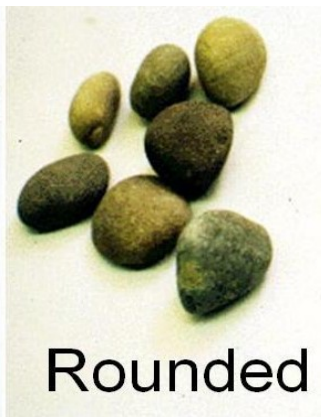
Flat Particles



Elongated Particles



Angular



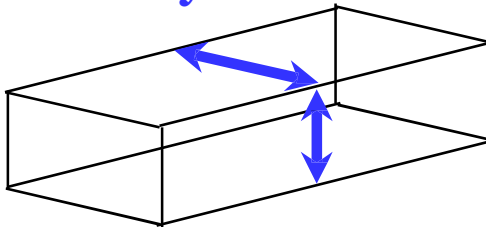
Rounded



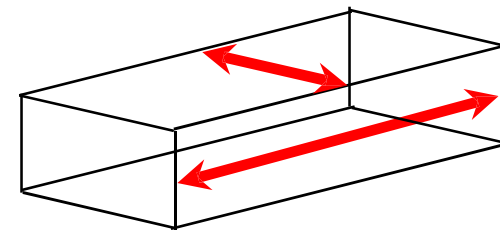
Particle Shape Characterization

- Equidimensional coarse aggregate particles are preferred
- Two types of particles that are not equidimensional:
 - Elongated
 - Flaky
- Particle is **flaky** if its smallest dimension (thickness) is less than 0.6 times of the “middle” dimension
- Particle is **elongated** if its largest dimension (length) is more than 1.8 times of the “middle” dimension

Flaky



Elongated



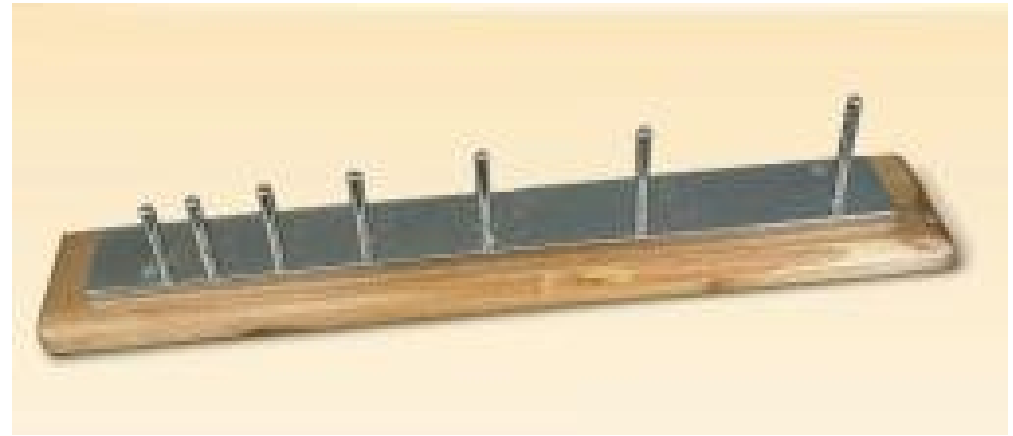


Flaky vs. Elongated



Flakiness Gauge

measures Flakiness Index



Length Gauge

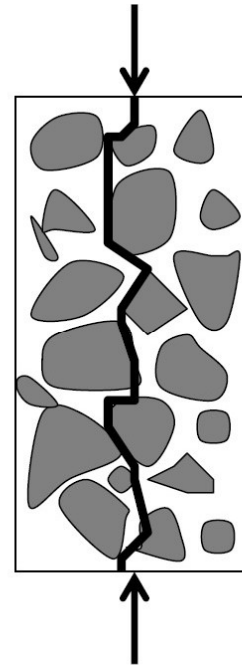
measures Elongation Index

- Flaky Particles: Thickness less than 0.6 of nominal size
- Elongated Particles: Length more than 1.80 times the nominal size

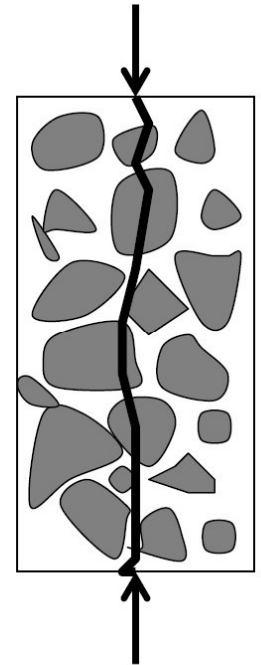
Failure Mode of Concrete in Compression



- Shape and surface texture of aggregate influence strength of concrete
- These factors are particularly important for high strength concrete
- Rougher texture results in larger adhesive force between particles & cement matrix
- Larger surface area of angular aggregate leads to larger adhesive force



Normal Strength Concrete



High Strength Concrete



Crushing Test of Bulk Aggregate

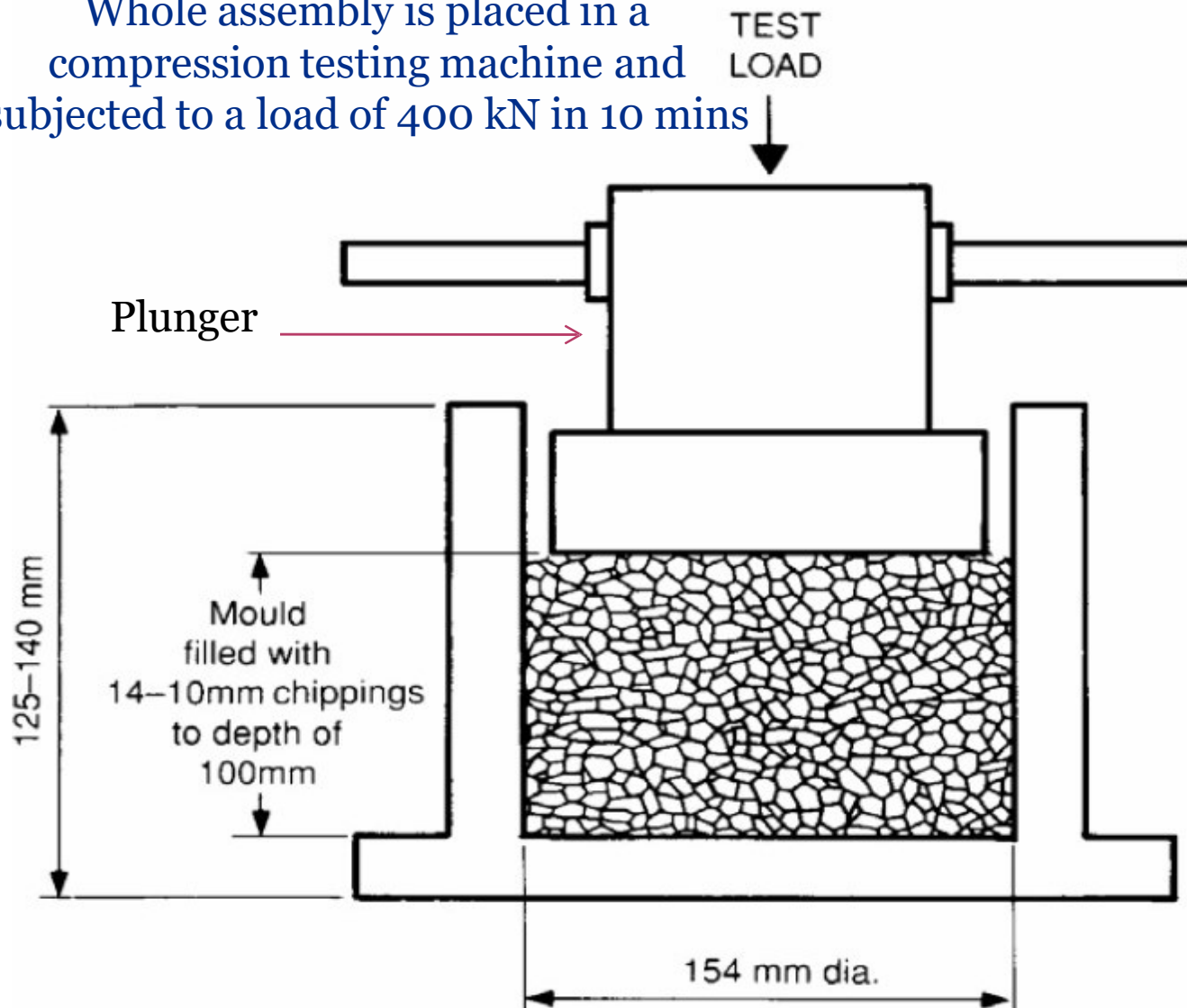
- Test on crushing properties of bulk aggregates: Crushing value test of BS 812: Part 110 : 1990
- Measures the resistance to pulverization
- Useful guide when dealing with aggregates of unknown performance
- There is no obvious physical relation between crushing value and compressive strength but results are usually in agreement
- Test rather insensitive to variation in strength of weaker aggregates - “Ten per cent fines value test” is introduced

$$\text{Crushing value} = \frac{\text{weight of material passing sieve}}{\text{total weight of sample}} \times 100\%$$



Crushing Test of Bulk Aggregate

Whole assembly is placed in a compression testing machine and subjected to a load of 400 kN in 10 mins



For normal aggregate

TEST LOAD

Aggregate Crushing Test
40 tonnes

10% Fines Test

A series of test loads between 40 tonnes and 1 tonne to interpolate the load which produces 10% of fines passing a 2.36mm sieve

For weak aggregate



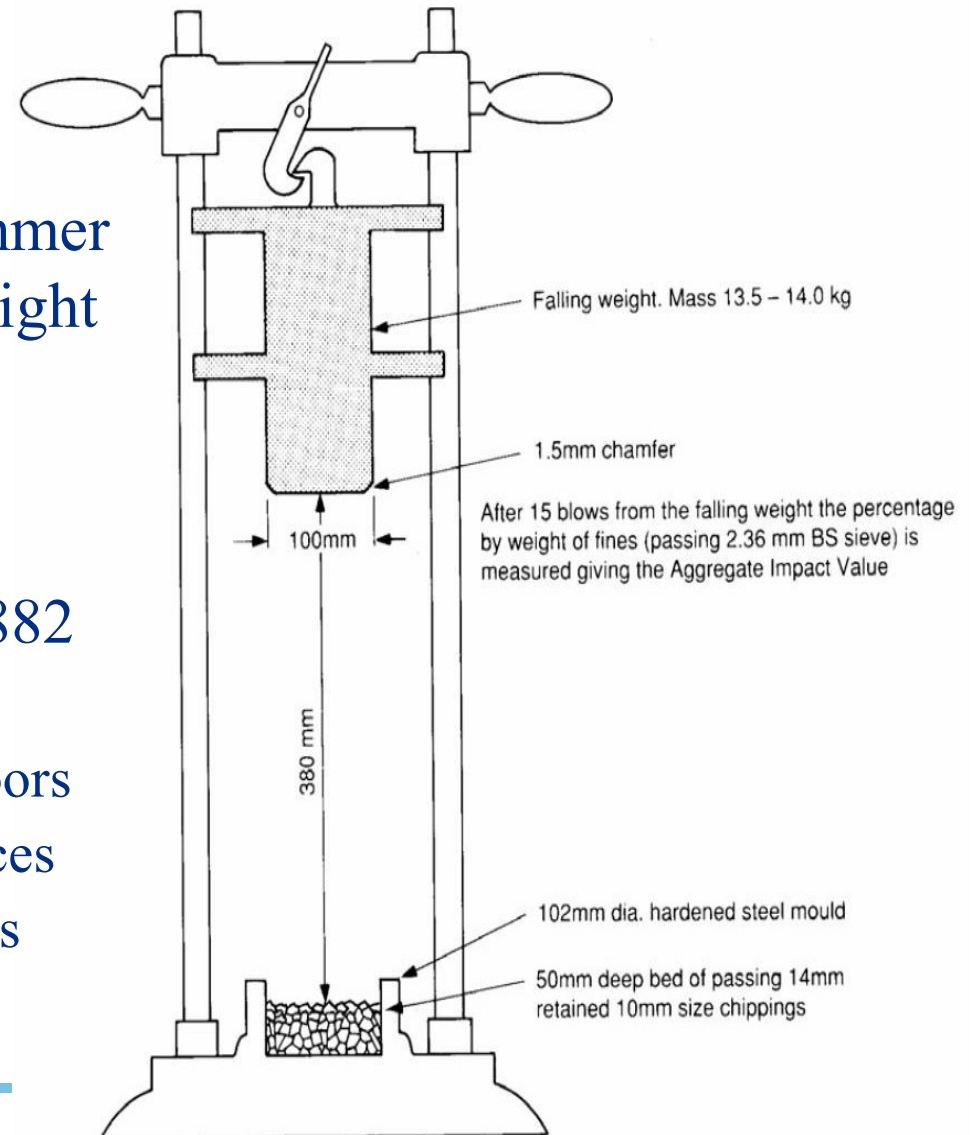
10% Fines Value Test

- Test included in BS 812 : Part 111 : 1990
- Apparatus of standard crushing test is used to determine load required to produce 10% of fines
- Achieved by applying a progressively increased load in 10 minutes
- Applied force results in a percentage of fines (7.5-12.5%) passing 2.36 mm (No. 8 ASTM) sieve
- Higher obtained load denotes a higher strength of aggregate
- BS 882 : 1992 prescribes minimum ten percent value of:
 - 150 kN (15 ton) - aggregate in heavy duty floor
 - 100 kN (10 ton) - aggregate in concrete for wearing surfaces
 - 50 kN (5 ton) - aggregate in other concretes



Impact Value of Bulk Aggregate

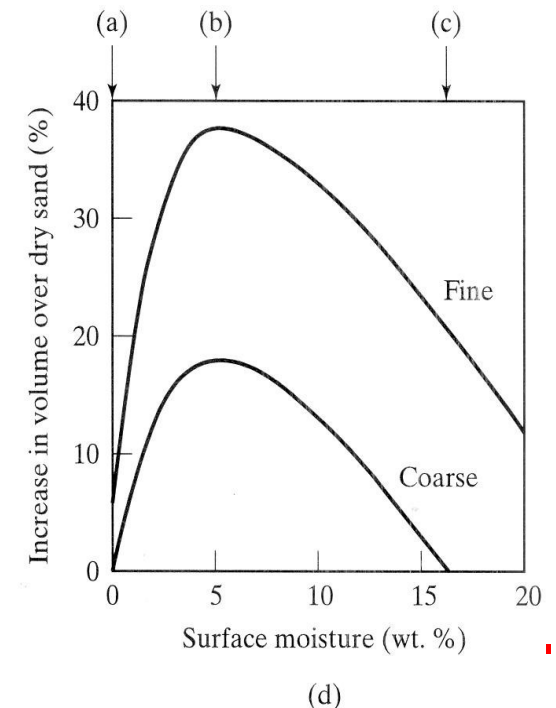
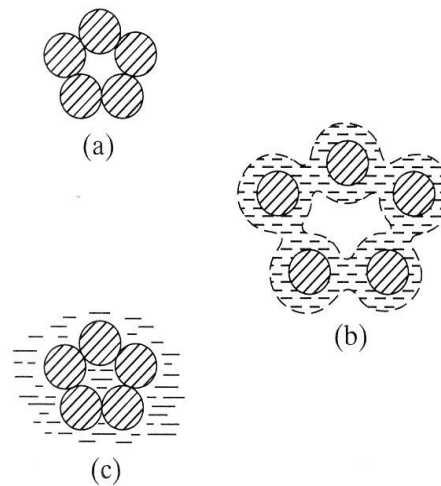
- **Toughness** is the resistance of a sample rock to failure by impact
- Impact applied by a standard hammer falling 15 times under its own weight upon the aggregate
- This results in fragmentation
- BS 812 : Part 112 : 1990 and BS 882 :1992 requires:
 - 25% for aggregate in heavy duty floors
 - 30% for aggregate in wearing surfaces
 - 45% for aggregate in other concretes



Bulking

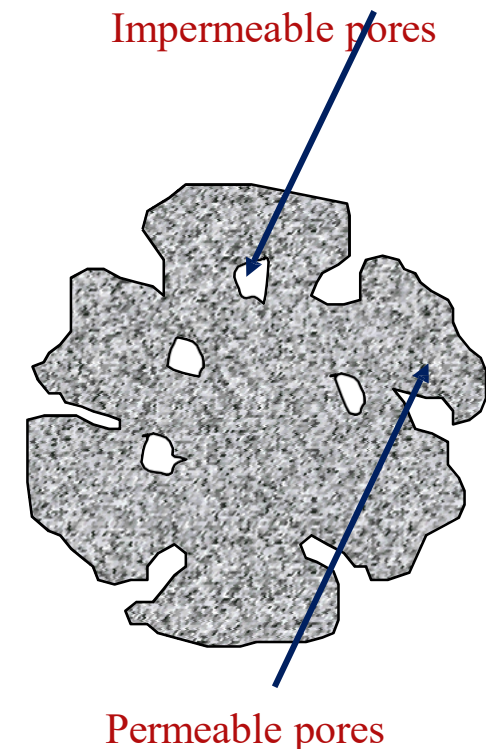


- Presence of moisture in sand creates bulking
- Bulking is the increase in volume of a given mass of sand caused by the films of water pushing the particles apart
- Formation of meniscus creates thicker water films between aggregate particles, pushing them apart and increasing the apparent volume
- Bulking can cause errors in mix proportioning by volume
- For that reason, aggregate is batched by weight
- Coarse aggregates show much less bulking since particle size is large compared to thickness of water film and effect of meniscus formation is slight



Effects of Voids in Aggregates

- A majority of natural aggregates have a SG between 2.6-2.7
- Permeable pores in aggregates create multiple definitions of specific gravity:
- **Absolute**: Volume of solid material excluding all pores
- **Apparent (ASG)**: Volume of solid includes impermeable pores but not capillary pores
- **Bulk-dry (BSG_{OD})** and **Bulk-saturated surface dry (BSG_{SSD})**: Since water contained in all pores of aggregate does not take part in chemical reactions of cement, it is considered as a part of aggregate

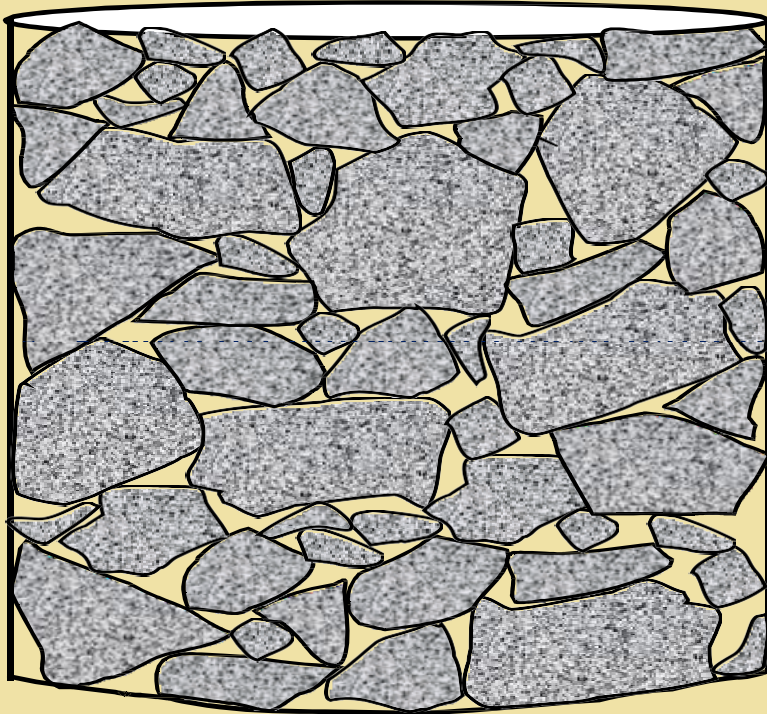




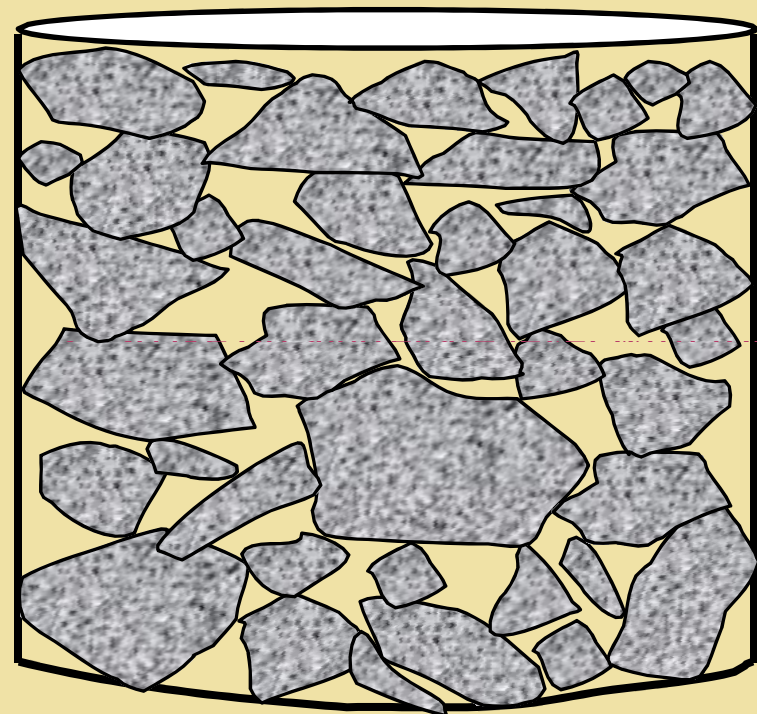
Bulk Unit Weight & Voids in Aggregates

- Sometimes we need to know the mass or weight of aggregate required to fill a volume (e.g. the volume of coarse aggregate in a cubic yard of concrete, for the proportioning of concrete):
- **Bulk unit weight** or bulk density is the weight of aggregate required to fill a “unit” volume. Typical units are cubic meters and cubic feet
- **Voids in aggregate** is the percentage of voids between aggregate particles

Test on Aggregate Bulk Unit Weight (ASTM C29)



AASHTO T19



AASHTO T19



Example

- Coarse aggregate is placed in a rigid bucket and rodded with a tamping rod to determine its unit weight. The following data are obtained:
 - Volume of bucket = 0.01 m^3
 - Weight of empty bucket = 8.4 kg
 - Weight of bucket filled with dry rodded coarse aggregate = 25.4 kg
- (a) Calculate the **dry-rodded unit weight**
- (b) If the oven-dried bulk specific gravity of the aggregate is 2.63 , calculate the **percent voids in the aggregate**

Cleanness and Deleterious Materials



- There are THREE main categories of deleterious substances in aggregates:

Impurities

- Organic impurities, clay, and other fine materials (e.g. silt and crusher dust)
- Interfere with cement hydration

Coatings

- Prevent the development of good bond between aggregate and cement paste

Individual Particles

- Produce weak or unsound sections
- Two types of unsound particles: (1) fail to maintain integrity, (2) cause disruptive expansion on freezing or water exposure

Cleanness and Deleterious Materials

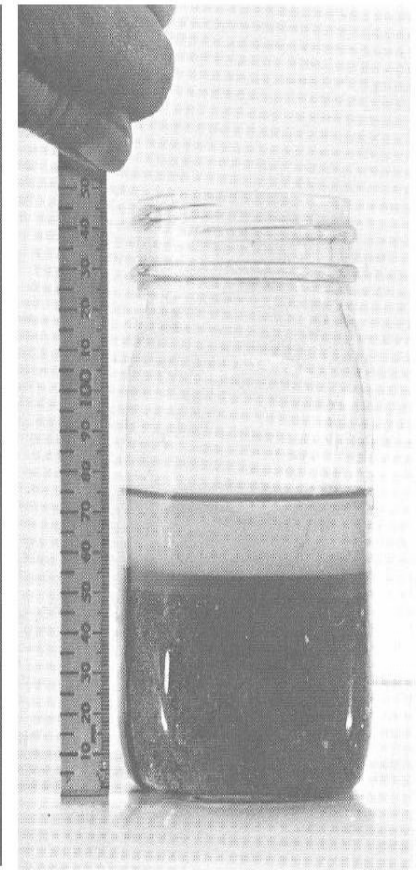
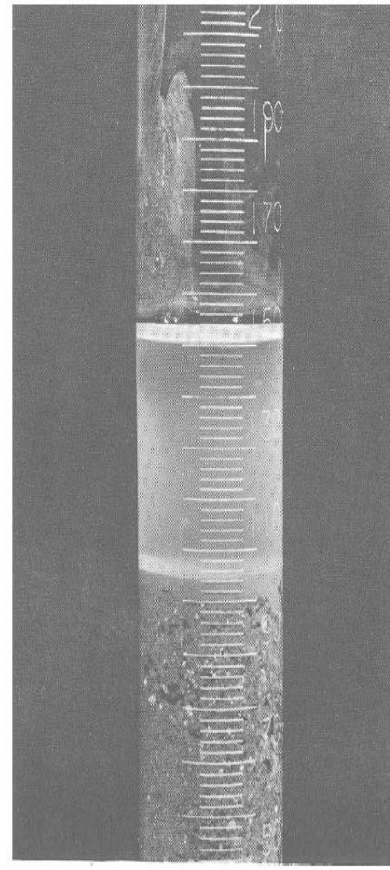
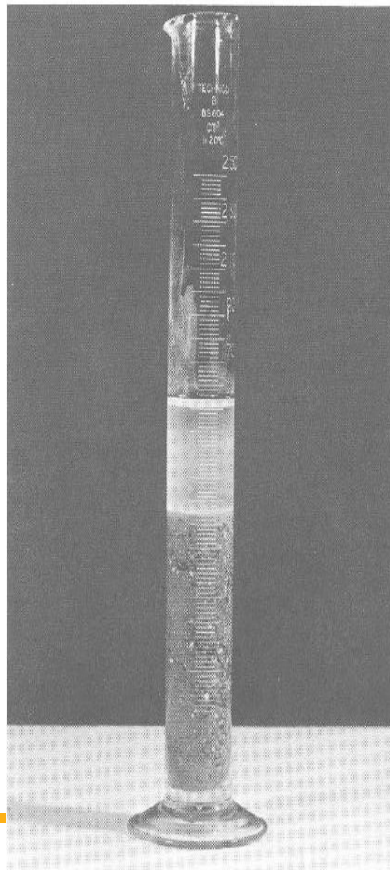
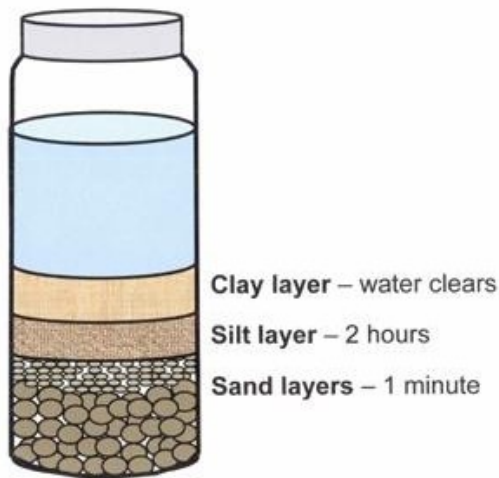


- **Deleterious Substances:** Aggregate contaminated by materials that adversely affect the quality of concrete

Substance	Harmful Effect
Organic impurities	Delay setting and hardening, may reduce strength gain, may cause deterioration
Minus 0.075 mm (No.200)	Weaken bond, may increase water requirement
Coal, lignite or other low-density materials	Reduce durability, may cause popouts or stains
Clay lumps and friable particles	Popouts, reduce durability and wear resistance
Soft particles	Reduce durability and wear resistance, popouts

Silt Test for Sand

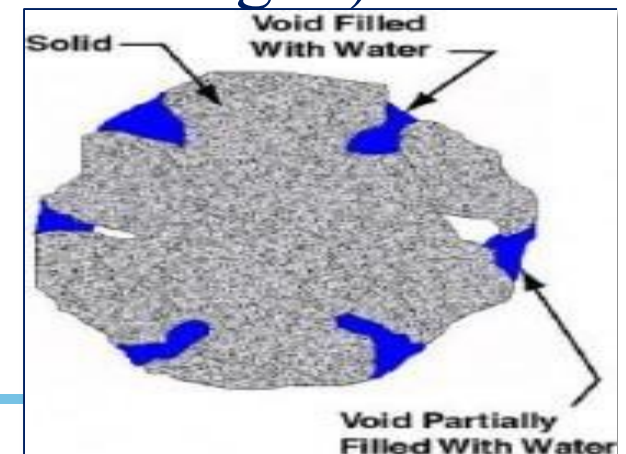
- Silt content = (height of silt layer / height of sand)
- Silt content should not be more than 8%





Soundness of Aggregate

- Soundness: Ability of aggregate to resist excessive changes in volume caused by changes in physical conditions
- Physical causes of volume changes: freezing and thawing, thermal changes, alternating wetting/drying
- Unsound aggregate: when volume changes cause deteriorations of concrete (e.g. pop-outs, cracking...)
- Tests for soundness of aggregate:
 - British - BS 812 : Part 121 :1989
 - American - ASTM C 88-90



Soundness & Durability

- Water freezes in voids and fractures and disintegrates aggregates
- Test method uses salt solution (sodium or magnesium sulfate) to simulate freezing (ASTM C88)
- Formation of salt crystals in aggregate pores disturbs particles
- Reduction of particle sizes denotes the degree of unsoundness



Prepare sample
minimum mass
specified gradation



Soak 16 hrs – dry 4 hrs
Repeat cycle 5 times



Measure gradation

$$Loss = \left(\frac{M_B - M_A}{M_B} \right) 100$$

Abrasion



- Wear of aggregate by abrasion through the:
 - rubbing of a foreign material against the stone under test (e.g. abrasion charge in a rotating drum)
 - attrition of stone particles against each other (e.g. friction between particles)
- Abrasion value test: BS 812 : Part 113 : 1990
- Aggregate particles between 10-14 mm tested
- Aggregate abrasion value: percentage loss in mass on abrasion
- Higher the abrasion value, lower is the resistance to abrasion
- Los Angeles Test (ASTM C131, C535):
 - Combines attrition and abrasion
 - Provides a good correlation with actual wear of aggregates and strength of concrete containing the aggregate

Los Angeles Test

- Aggregate of specified grading is placed in cylindrical drum, mounted horizontally with a shelf inside
- Charge of steel balls is added
- Drum is rotated a specified number of revolutions (e.g. 500)
- Tumbling and dropping of the aggregate and balls result in abrasion and attrition of the aggregate





Hardness & Abrasion Resistance

▫ Resist load damage

- During construction
- Traffic loads



- Prepare sample
- Minimum original mass
- Specified gradation

▫ Los Angeles abrasion test (ASTM C131, C535)



- Charge drum w/ sample
- Steel spheres
- 500 revolutions



• Sieve

$$Loss = \left(\frac{M_{original} - M_{final}}{M_{original}} \right) \times 100$$

Alkali-Aggregate Reactivity

- **Alkali-silica reaction:** Silica in some aggregate reacts with the alkalis (Na_2O , K_2O) in Portland Cement (especially in warm, humid climates)

- Excessive expansion



- Cracking



- Spalling





Alkali-Aggregate Reactivity

- **Alkali-carbonate reaction:** Carbonates in aggregate can also react to a lesser extent
- If a reactive aggregate has to be used, reactivity must be minimized:
 - Type II cement – minimizes alkali content of PC
 - Keep concrete as dry as possible
 - Fly Ash, GGBS, silica fume reduce alkali reactivity
 - Sweetening – add 30% crushed limestone to the aggregate

Tests on Alkali-Aggregate Reactivity



- **ASTM C227** – tests expansion potential of cement-aggregate combination
 - expansion of mortar bar at specific temperature & humidity
- **ASTM C289** – reactive silicates in aggregate
- **ASTM C586** – reactive carbonates in aggregate

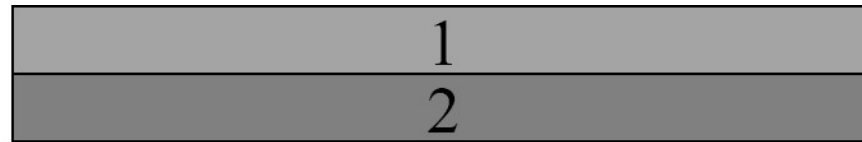
Thermal Properties of Aggregate



- There are **three** thermal properties of aggregate that may be significant in the performance of concrete:
 - Coefficient of thermal expansion
 - Specific heat
 - Conductivity

→ important in mass concrete
- If the coefficient of thermal expansion of coarse aggregate and of the hydrated paste differ too much:
 - a large change in temperature may introduce differential movement, and
 - break the bond between aggregate particles and surrounding paste

Thermal Expansion of Composite Materials



$T = 20^{\circ}\text{C}$

If $\alpha_1 \ll \alpha_2$





Thermal Properties of Aggregate

- The linear coefficient of thermal expansion:
 - Aggregates: $\sim 5 - 13 \times 10^{-6}$ per $^{\circ}\text{C}$
 - Hydrated cement paste: $\sim 11 - 16 \times 10^{-6}$ per $^{\circ}\text{C}$
- When the two coefficients differ by more than 5×10^{-6} per $^{\circ}\text{C}$, the durability of concrete can be affected
- A serious difference in coefficients occurs only with aggregates of a very low expansion (e.g. certain granites, limestones, and marbles)